Program: Information Technology **S.Y B.Tech Semester: IV**

Course: Python Laboratory (DJS23IPC253L)

Pre-requisite:
1. Structured Programming Approach
2. Java Programming

Course Objectives: The objective of the course is to expose students to a new programming language "Python3", thereby making them familiarized with the sequence data types and their interoperability, various control structures and object-oriented programming in Python. To enable students, to develop GUI Applications and Web Applications with database connectivity. To lay the foundation of Machine Learning and Data Science Techniques with visualizations

Course Outcomes: On successful completion of this course, student should be able to:
1. Develop python programs to solve real-world problems.
2. Apply different visualizations on a given dataset.
3. Create GUI applications using Flask.
4. Work effectively as a member of a team.

Detailed Syllabus: (unit wise)

Unit	Description	Duration
1	**Basics of Python**: Numbers in Python, Basic & Built-in Math functions, Number Formats, Strings, Quotes, print () Function, range() function, Assigning Values to Names & Changing Data Through Names, Copying Data, Tuples, Lists, Dictionaries, Sets, Numpy Arrays, Strings	04
2	**Control Statements and Functions:** If statement, if-elif-else, Repetition using while loop, for loop, Defining a Function, Checking & Setting Your Parameters, Default arguments, Variable length arguments, Defining and calling functions within a function, Layers of Functions, Lambda and Filter, Zip(), Map(), Reduce() function, recursion, Function Decorators	04
3	**Object Oriented Programming:** Creating a Class, Self-Variables, Constructors, Types of Methods, Constructors in Inheritance, Polymorphism, the super () Method, Method Resolution Order (MRO), Operator Overloading, Method Overloading & Overriding, Interfaces in Python. **Exceptions Handling**: Exceptions, Exception Handling, Types of Exceptions, The Except Block, The assert Statement, User Defined Exceptions	06
4	**Introduction To Data Science Packages:** Creating Modules and Packages, Documenting & Viewing Module, Basics of Testing Your Modules and Packages, Importing & exporting Modules, Random, Matplotlib, Pandas, SciPy, scikit learn Modules	06
5	**Files Handling:** Types of Files in Python, opening a File, closing a File, Writing Text Files, File content manipulation, working with Binary Files, Appending Text to a File, Reading Text Files, File Exceptions, The with Statement Pickle in Python, Zip File Module.	02
6	**GUI Programming with Database Connectivity:** GUI Programming Toolkits, Creating GUI Widgets with Tkinter, Creating Layouts, Form Components, Dialog Boxes. Types of Databases Used with Python, MySQL database Connectivity with Python, Performing DML operations on database. **Web Development Framework:** Flask -Templates, Flask Template Engine: Jinga, Flask-Jinga, Template creation, Rendering a web-based application.	06

Prepared by Checked by Head of the Department Principal

List of Tutorials/ Experiments:

1. Write python programs to understand Expressions, Variables, Quotes, Basic Math operations.
2. Write a Python program to implement Basic String Operations & String Methods.
3. Write a Python program to implement functions of List, Tuples, Dictionaries.
4. Write a Python program to implement Arrays / Numpy Array (1D, 2D) applications.
5. Write python programs to demonstrate applications of different decision making statements.
6. Write a Python program to implement Functions and Recursion.
7. Write a Python program to implement Programs based on Lambda, Map, Reduce Functions.
8. Write a Python program to implement program to implement concept of Function decorators.
9. Write python programs to implement Classes & objects, Constructors
10. Write python programs to implement Inheritance & Polymorphism.
11. Write a Python program to implement data analysis using pandas.
12. Write a Python program to implement basic scientific operations using scipy.
13. Write a Python program to implement data visualizations using matplotlib.
14. Write python programs to implement Exception handling.
15. Write python programs to understand different File handling operations with exception handling.
16. Write python programs to understand GUI designing and database operations (Programs based on GUI designing using Tkinter, Mysql database creation & Database connectivity with DML).
17. Write a Python program to implement Web based application with Flask Framework.

Books Recommended:

Text books:

1. Dr. R. Nageswara Rao, "Core Python Programming", Dreamtech Press, Wiley Publication
2. M. T. Savaliya, R. K. Maurya, "Programming through Python", StarEdu Solutions.
3. E Balagurusamy, "Introduction to computing and problem-solving using python", McGraw Hill Publication.

Reference Books:

1. Zed A. Shaw, "Learn Python 3 the Hard Way", Zed Shaw's Hard Way Series.
2. Martin C. Brown, "Python: The Complete Reference", McGraw-Hill Publication.
3. Paul Barry, "Head first Python", 2nd Edition, O'Reilly Media, Inc.

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